



The “covariation method” for estimating the parameters of the standard Dynamic Energy Budget model I: Philosophy and approach

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ABSTRACT

The Dynamic Energy Budget (DEB) theory for metabolic organisation captures the processes of development, growth, maintenance, reproduction and ageing for any kind of organism throughout its life-cycle. However, the application of DEB theory is challenging because the state variables and parameters are abstract quantities that are not directly observable. We here present a new approach of parameter estimation, the covariation method, that permits all parameters of the standard Dynamic Energy Budget (DEB) model to be estimated from standard empirical datasets. Parameter estimates are based on the simultaneous minimization of a weighted sum of squared deviations between a number of data sets and model predictions or the minimisation of the negative log likelihood function, both in a single-step procedure. The structure of DEB theory permits the unusual situation of using single data-points (such as the maximum reproduction rate), which we call “zero-variate” data, for estimating parameters. We also introduce the concept of “pseudo-data”, exploiting the rules for the covariation of parameter values among species that are implied by the standard DEB model. This allows us to introduce the concept of a generalised animal, which has specified parameter values. We here outline the philosophy behind the approach and its technical implementation. In a companion paper, we assess the behaviour of the estimation procedure and present preliminary findings of emerging patterns in parameter values across diverse taxa.

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1. Introduction

Metabolic theories aim to capture how organisms acquire energy and matter from their surroundings and allocate it to growth, maintenance, development and reproduction. These processes subsequently affect a wide range of fundamental biological phenomena such as ageing and body size. These individual-level processes and outcomes show enormous variation across species and are well known to be reflected in patterns at higher levels of biological organisation including communities and ecosystems. Mechanistic theories of metabolism that can capture this variation should therefore provide a powerful basis from which to answer a very wide range of problems in ecology and evolutionary biology.

The Dynamic Energy Budget (DEB) theory is such a mechanistic theory that is unique in its generality and comprehensiveness (Kooijman, 2010; Sousa et al., 2008, 2010). It captures the metabolic dynamics of an

individual organism through its entire life-cycle, be it ectothermic or endothermic, autotrophic or heterotrophic, and is explicitly tied to food/substrate availability and temperature. DEB theory is not only very general but also surprisingly simple. The simplicity and generality of DEB theory comes at a cost, however, in that its formulation involves a high level of abstraction. For example, total body mass is abstracted into the concepts of “reserve” and “structure”, which are primary state variables but cannot be directly observed. The parameters of the model have dimensions that relate to the state variables and thus are themselves difficult or impossible to observe directly. While this is a common circumstance in a variety of fields employing formal mathematical models (who has ever seen a sub-nuclear particle?), it poses challenges for applying the theory and for comparative analysis. The concept molecule was developed long before any molecule was actually seen. A similar situation might also apply to metabolic variables as defined by DEB theory: variables that are presently abstract and measurable indirectly only can become less abstract in the future and directly measurable.

For many applications of DEB theory, not all parameters need to be known and such a subset is sufficient in many situations. If our data does not include quantities with energy in their units, for instance,

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but our model predictions also have this property, we know on dimensional grounds already, that we don't need to know any parameter with energy in its units. We only need to know simple functions of those parameters, such that the dimension energy drops out. Knowledge about such functions is weaker than knowledge about each parameter value. Thus far, most applications of DEB theory have involved estimation of only a subset of parameters (e.g. Kooijman, 1988; Zonneveld and Kooijman, 1993, but see Cardoso et al., 2006; Jager et al., 2004; Veer et al., 2006, 2010). These previous estimation procedures have taken the strategy of estimating “compound parameters” that represent amalgams of the core DEB parameters that are more easily estimated from typical empirical data sets (Kooijman et al., 2008; Meer, 2006).

However, for many applications, such as comparative analyses of evolutionary trends in physiological traits and ecological or geographical constraints on abundance, we need *all* parameters, since only then we can evaluate the full energy and mass budgets dynamically. The present paper, and its companion (Lika et al., 2011-this issue), introduces a new method to access all core parameters of the DEB model simultaneously from a collection of common empirical measurements. We develop the approach for the standard DEB model, which is the simplest non-degenerated model that is implied by DEB theory; the ageing module is also included to allow for examination of implied properties at the population level. In principle, however, it can be generalised to more complex DEB models, e.g. that involve changes in shape during growth or multiple structures and reserves. A key feature of the method is that it exploits the rules for the covariation of parameter values, as implied by the physical laws on which DEB theory has been built.

Since we might arrive at a good fit for the wrong reasons, we wrote software <http://www.bio.vu.nl/thb/deb/deblab/debtool> to predict some 100 quantities on a wide variety of aspects of the DEBs, given parameter values. The file `pars_my_pet` in the `add_my_pet`-collection generates these predictions. The idea is to inspect this list critically and, in the case of unlikely predictions, add a likely value to the list of observations and repeat the parameter estimation procedure till all inconsistencies are resolved, see Appendix A. We hope that this procedure will guide experimental research. We see this check as an essential step in an estimation procedure, where physical, chemical and biological realism has priority over goodness of fit. Where the classic approach in statistics is to assume that the model is true and the observations suffer from noise, we here want to fit a model which we know is very simple, given the immense complexity of living organisms. We know a priori that the model is wrong in the strict sense of the word, but possibly useful.

In this paper, we first describe the structure, state variables and parameters of the standard DEB model and their relationship to empirical observations. We then outline the covariation method to estimate parameters from data. In the companion paper, we study the properties of the method and present the beginnings of a parameter library for diverse species and the initial patterns that emerge from it (Lika et al., 2011-this issue). While the present method only exploits the covariation of parameter values implied by physical laws, future developments might also exploit evolutionary and ecological patterns to reduce the need for data and increase efficiency and reliability.

2. Relating variables to observations

To understand how the abstract parameters of the DEB theory can be estimated from observations in the real world, it is necessary to provide a brief summary of the structure of DEB theory and the different types of parameters and variables. The standard DEB model applies to an isomorphic organism (no change in shape during growth). Knowledge of the parameters of the DEB model allows predictions of changes in the state variables of structure, reserve, reproduction buffer and maturity through ontogeny under any combination of food

levels and body temperature. This gives a complete, stoichiometrically constrained picture of fluxes of energy and mass as the organism grows, develops, reproduces and ages throughout its life cycle in dynamic environments (temperature, food availability). The standard DEB model achieves this using a single food type, a single reserve and a single structure, all with a constant chemical composition (although these components may differ in composition from each other).

2.1. Core vs. auxiliary parameters

In DEB theory, a distinction is made between “core” parameters and “auxiliary” parameters. This distinction is critical to the ideas behind DEB parameter estimation, as illustrated in Fig. 1. The core DEB parameters (Table 1) are intimately linked to the underlying assumptions of DEB theory and relate directly to processes controlling state variable dynamics. These processes include life cycle transitions (embryo, juvenile and adult: E_H^b , E_H^j), feeding ($\{\dot{F}_m\}$), digestion (κ_X), assimilation ($\{\dot{p}_{Am}\}$), mobilisation ($\dot{v}$), allocation (κ), maintenance ($\{\dot{p}_M\}$, $\{\dot{p}_T\}$), growth ($[E_G]$), reproduction (κ_R), development ($\dot{k}_j$), respiration (dioxxygen consumption, dissipating heat), product formation (faeces (κ_X^f), nitrogen waste, carbon dioxide, water) and ageing ($\dot{h}_a$, s_G). All aspects of the fluxes of energy and mass (i.e. chemical elements) are covered in dynamic environments (temperature, food availability), including stoichiometric constraints on production, while biomass can change in chemical composition.

While core DEB theory deals with the dynamics of abstract variables, including the effects of temperature and food availability, auxiliary theory links these variables to quantities that can be measured directly such as length, wet or dry weight, respiration, egg output, etc. Auxiliary theory involves additional “auxiliary parameters”, see Table 2. This theory specifies, for instance, the relationships between volumes, wet and dry weights, energies and other measurable quantities in the context of DEB theory. The combination of core and auxiliary theory defines mapping functions from the abstract world to the real world and vice versa. Auxiliary parameters are less intimately linked to the concepts on which the model is based, but they still rest on some underlying principles, such as strong homeostasis. Fig. 1 illustrates the logical position between the various quantities.

Both core and auxiliary theories simplify and idealise a complex reality substantially, and they do that on the basis of a list of explicit assumptions that, in combination, fully specify the theory (see Table 2.4 in (Kooijman, 2010) for a complete list of the assumptions that specify the standard DEB model quantitatively and chapter 3 for the auxiliary theory).

2.2. Primary vs. compound parameters

Another important distinction is made in DEB theory between “primary” and “compound” parameters. A primary parameter is most intimately connected to a single underlying process, while compound parameters (i.e. simple functions of parameters) typically depend on several underlying processes. The listed core and auxiliary parameters in Tables 1 and 2 are considered as primary parameters, on which compound parameters can be based, such as the core parameters maximum reserve capacity $[E_m] = \{\dot{p}_{Am}\}/\dot{v}$, J cm^{-3} and maximum structural body length $L_m = \kappa\{\dot{p}_{Am}\}/[\dot{p}_M]$, cm and the auxiliary parameter volume-specific structural mass $[M_V] = d_V/w_V$, mol cm^{-3} .

2.3. Real-data vs. pseudo-data

The covariation method aims to estimate the core DEB parameters from real data, i.e. empirical observations of physiological processes. However, the estimation procedure may be guided by prior knowledge of parameter values, not unlike the concept of a “prior” in Bayesian parameter estimation methods. Conceptually, this prior knowledge is

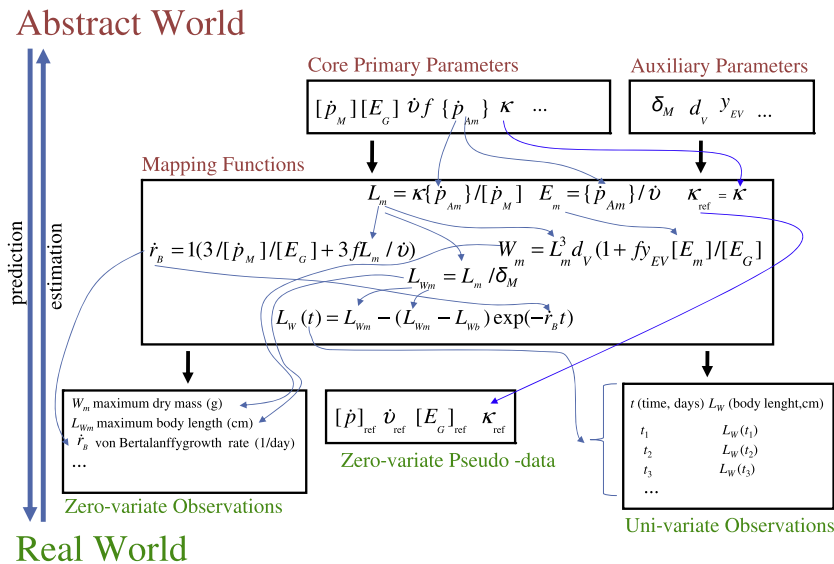


Fig. 1. A schematic depiction of the linkages between the abstracted DEB theory constructs and parameters and real world observations. Mapping functions comprising primary and auxiliary parameters can be constructed for real world observations that may consist of single values (zero-variate data) or arrays of observations, e.g. through time (uni-variate data). These mapping functions are defined by core and auxiliary DEB theory. Only a subset of DEB parameters and processes are indicated. Note that a given DEB primary parameter can be connected to multiple real world observations. For DEB parameter estimation, this has the consequence that an increasingly wide array of real world observations provides constraints on possible parameter values. These linkages between DEB parameters and empirical observations can be considered metaphorically in terms of the genotype–phenotype map. The DEB parameters can be considered as genes that interact via core and auxiliary DEB theory (developmental transformation) to produce observations that can be considered as phenotypes. Environmental forcing (food, temperature) produces reaction norms such that the same parameter set expresses different empirical outcomes. The DEB parameters can be considered as highly ‘pleiotropic’ in action, such that a given parameter has multiple ‘phenotypic’ effects. Pseudo-data are parameter values corresponding to a generalised animal, i.e. typical values for a wide variety of animals. These parameter values have been tested to have biologically and physically acceptable implied properties. Estimated parameter values should always be assessed in this way, since physical realism has priority over goodness of fit. Parameter values with a poorer fit may have much more physical realism, due to the fact that observations do not include all possible aspects of energy budgets. The covariation method includes such assessments.

treated as data and hence we refer to it as “pseudo-data”. The role of pseudo-data in the covariation method will be described further below.

2.4. The rules for parameter covariation and their relation to body-size scaling patterns

The basic philosophy of DEB theory is to access the parameters for many individuals of many species independently (in principle) and plot them in the parameter space, giving one point per individual. We expect that the points of individuals that belong to the same species tend to cluster. The covariation rules tell how these parameter values are distributed in this space, based on plain physics only. The reasoning does not include any selective pressures acting independently on parameter values. The covariation rules can thus be considered as a “null expectation” for parameter covariation and their impact on the related physiological quantities. Evolutionary and ecological adaptation can cause deviations from these physical expectations, so these deviations can guide us to specific adaptations. The popular approach of body size scaling relationships might look similar at first sight, but we see important differences with what we try to do. Where body size is treated in such relationships as an independent variable and eco-physiological properties are regressed empirically as (allometric) functions of body size, body size in the context of DEB theory is a result of underlying processes, is not considered as an independent variable and is affected by environmental variables. DEB theory also refrains from empirical relationships and, thus, does not make use of allometric functions. It is then possible to write physiological quantities, such as maximum body weight, minimum incubation time, or maximum reproduction or respiration rate as functions of parameters. These expressions can then be rearranged so that a physiological process, e.g. respiration, is a function of body weight.

DEB theory sharply distinguishes between intra-species comparisons (different body sizes, same parameters) and inter-species comparisons (different body sizes but also different parameters). This essential distinction is typically absent in many body size scaling

relationships in the literature. The specific respiration of a small young mouse is higher than that of a fully grown one, because the young one grows. The specific respiration of a fully grown mouse is higher than that of a fully grown elephant, but obviously for a very different reason, which is well-understood in the context of DEB theory (Kooijman, 1986). A large collection of eco-physiological quantities (Kooijman, 1986, 2010) was found to be highly consistent with empirical inter-species body size scaling patterns.

Although observed physiological quantities might depend on maximum body weight in rather complex ways, the covariation structure of the primary parameters is very simple. The rules for the covariation of parameter values, which we exploit in the present parameter estimation method, result from a combination of three conditions:

1. all parameters have a clear physical meaning and can, therefore, be classified as intensive or design parameters
2. appropriate ratios of design parameters are intensive (in view of their physical meaning)
3. maximum structural length depends on a single design parameter (so no degree of freedom to link parameter values to length).

The first condition classifies DEB theory as mechanistic by definition. The last condition ensures that the maximum specific assimilation rate, $\{\dot{p}_{Am}\}$, is proportional to the maximum structural length. The second condition links all other parameters to each other. Notice that the rules for the covariation of parameters follow from core theory and are therefore fully independent of auxiliary theory and of empirical observations. These rules are implemented in the parameter estimation procedure using the concept of pseudo-data described above, in that only intensive parameters can play the role of pseudo-data points.

The covariation rules can also be useful for making inter-specific comparisons of parameter values because it allows all organisms to be normalised to a common maximum (structural) body length. The comparison is done on the basis of a reference maximum structural body length of $L_m^{ref} = 1$ cm using the zoom factor $z = L_m/L_m^{ref}$, where

Table 1

The core parameters of the standard DEB model. Symbol, typical value at $T_{\text{ref}} = 293$ K and formal range limits are indicated.

Zoom factor $z = L_m/L_m^{\text{ref}}$ with $L_m^{\text{ref}} = 1$ cm. Controls the maximum length $L_m = \kappa\{\dot{p}_{Am}\}/[\dot{p}_M]$ via the specific assimilation $\{\dot{p}_{Am}\}$. It also affects the life-stage parameters and the ageing acceleration as primary parameters. Most derived quantities (being functions of parameters) are affected by this factor in ways that follow from DEB theory. In combination with the zoom factor, the shape coefficient δ_M controls the length–weight relationship, where weight has contributions from reserve and, possibly, the reproduction buffer. An increase in the zoom factor gives an increase in the specific assimilation rate and so an increase in the maximum reserve capacity and in the contribution of reserve to weight. Range: >0 ; likely range <150 . Maximum surface-area-specific searching rate $\{\dot{F}_m\}$, $65 \text{ dm}^3 \text{ d}^{-1} \text{ cm}^{-2}$: controls food intake if food is not abundant and has no effect at abundant food; its value and even its dimension is species-specific; for some taxa the unit $\text{dm}^2 \text{ d}^{-1} \text{ cm}^{-2}$ is more appropriate. A low value means a low food density at which food starts to limit ingestion. Range: >0 . Digestion efficiency κ_X , 0.8: specifies the fraction of energy in food that is fixed in reserve. Since this factor is applied to the maximum specific assimilation to obtain the maximum specific food intake, a low value for κ_X gives a high food intake. Range: >0 , <1 . Defecation efficiency κ_X^p , 0.1: specifies the fraction of energy in food that ends up as faeces. This parameter does not affect the state variables (structure, reserve, maturity, damage), but controls faeces production and mineral fluxes (CO_2 , H_2O , O_2 , NH_3). Notice that $\kappa_X + \kappa_X^p < 1$, else energy (and carbon dioxide) uptake will occur directly from the environment. Range: >0 , <1 . Energy conductance $\dot{v}$, 0.02 cm d^{-1} : controls the reserve mobilisation. A high value gives a high growth rate, short development time to reach birth or maturity, a low maximum reserve density, a rapid occurrence of problems during starvation. The maximum size is not affected by this parameter. The energy conductance is unlikely to deviate strongly from the typical value independent of the body size of the species. Range: >0 . Allocation fraction to soma κ , 0.8: controls the allocation of mobilised reserve to somatic maintenance and growth as opposed to maturity maintenance and maturation of reproduction. A high value gives rapid growth to a large size, long development times and low reproduction. A small value ($\kappa < 0.5$) can reduce growth and reproduction, since food uptake is linked to size. Range: >0 , <1 . Reproduction efficiency κ_B , 0.95: is the fraction of reserve allocated to reproduction that is fixed in the reserve of offspring. A high value gives a high reproduction rate and a low CO_2 production that is linked to reproduction. Range: >0 , <1 . Volume-specific somatic maintenance $[\dot{p}_M]$, $18 \text{ J d}^{-1} \text{ cm}^{-3}$: controls the sink of reserve linked to structural volume, mostly due to turnover of structure and behaviour, transport. A high value reduces growth as well as the maximum size, so indirectly also reproduction. This effect can be cancelled by increasing the zoom factor. Range: >0 . Surface-specific somatic maintenance $\{\dot{p}_T\}$, $0 \text{ J d}^{-1} \text{ cm}^{-2}$: also controls the sink of reserve, but now linked to structural surface area. Heating in endotherms and osmotic work in freshwater organisms are examples of such sinks. It thus depends in environmental temperature and salinity, respectively. As long as endotherms are in the thermal-neutral zone, $\{\dot{p}_T\} = 0$. A high value reduces the ultimate length, but the von Bertalanffy growth rate does not depend on this rate. Range: ≥ 0 . Maturity maintenance rate coefficient k_j , 0.002 d^{-1} : is the third primary parameter that controls the sink of reserve, but now linked to maturity. Metabolic switches (birth, puberty) occur at threshold values for maturity. A high value delays development and reduces reproduction. Maturity density (maturity per structure) is constant if $k_j = [\dot{p}_M]/[E_C]$. Range: >0 ; likely range $<[\dot{p}_M]/[E_C]$. Specific cost for structure $[E_C]$, 2800 J cm^{-3} : is the reserve energy that is required to synthesise a unit volume of structure. It includes the energy content of the tissue plus the overhead costs of the anabolic machinery. A high value reduces the growth rate (but not the ultimate size), and decreases the size at birth (and puberty). If toxicants increase this value, hormesis occurs for reproduction (i.e. low levels of toxicants increase the reproduction rate). The value is proportional to the specific density d_v for dry mass (in g cm^{-3}). Range: $>\mu_v[M_v] = \mu_v d_v/w_v$. Maturity at birth E_H^b , $0.275z^3 \text{ J}$: controls the timing of and the size at birth, i.e. the moment assimilation is switched on. Food abundance affects the timing, because of the maternal effect (reserve density at birth equals that of the mother); eggs with little reserve take longer to develop into a hungry neonate. Increasing the value causes an increase in the size at birth and a decrease of the reproduction rate. Range: >0 , $<\{\dot{p}_{Am}\}^3 [E_C] (1-\kappa) \kappa^2 / [\dot{p}_M]^3 = L_m^3 [E_m] g (1-\kappa)$. Maturity at puberty E_H^p , $166z^3 \text{ J}$: controls the timing of and the size at puberty, i.e. the moment at which investment into maturation is re-directed to reproduction. Food abundance affects the timing sensitively. Increasing the value causes an increase in the size at puberty. Range: >0 , $<\{\dot{p}_{Am}\}^3 [E_C] (1-\kappa) \kappa^2 / [\dot{p}_M]^3 = L_m^3 [E_m] g (1-\kappa)$; likely range $>E_H^b$. Weibull ageing acceleration $\ddot{h}_a$, $10^{-6} z \text{ d}^{-2}$: controls the mean life span in a way that hardly depends on food density (because the increased respiration is cancelled by dilution by growth). Increasing the value reduces the mean life span and the survival probabilities at birth and puberty. Range: >0 .

Table 1 (continued)

Gompertz stress coefficient s_G , 0.01: also controls the mean life span, but in ways that depend on food density. A positive value (around 0.5) can be expected in endotherms and elongates life at caloric restriction. Negative values can occur if damage inducing compounds can be degraded (found for *Daphnia magna*). Increasing the value decreases the mean life span, but increases survival at young age, relative to the mean life span. Range: ≥ 0 , $<\infty$; likely range ≥ 0.5 , <0.5 .

the maximum structural body length $L_m = \kappa\{\dot{p}_{Am}\}/[\dot{p}_M]$ depends on 3 parameters: the allocation fraction to soma κ , the maximum specific assimilation rate $\{\dot{p}_{Am}\}$ and the specific somatic maintenance rate $[\dot{p}_M]$. Likewise a comparison of a particular rate or time for two species is mostly done on the basis of the same (body) temperature $T_{\text{ref}} = 293$ K.

3. The covariation method of parameter estimation

Taking the concepts described above of core and auxiliary parameters, together with the rules for covariation, we can now describe the

Table 2

The conversion and temperature and scaling parameters of the standard DEB model. Symbols, units and formal range limits are given.

Conversion parameters

Shape coefficient δ_M : converts physical to volumetric structural length; its value is meaningless without definition of how the length is taken (e.g. snout to vent). In combination with the zoom factor, the shape coefficient controls the length–weight relationship, where weight has contributions from reserve. Range: >0 . Likely range <0.2 for snout-to-tail-tip

Specific densities $d_O = (d_X \quad d_V \quad d_E \quad d_P)$, g cm^{-3} : converts volumes to grammes for food X , structure V , reserve E , and faeces P . They relate the frequency of the chemical elements H, O and N to C and differ between wet and dry mass; the values of the specific densities must be consistent with the choice made for the chemical indices; if water is included in the chemical indices, it must also be included in the specific densities. Typical value: $d_O = (0.1 \quad 0.1 \quad 0.1 \quad 0.1) \text{ g cm}^{-3}$, but see text.

Chemical potentials $\mu_O = (\mu_X \quad \mu_V \quad \mu_E \quad \mu_P)$, J mol^{-1} : convert moles to energy for food X , structure V , reserve E , and faeces P . $\mu_M = (\mu_C \quad \mu_H \quad \mu_O \quad \mu_N)$, J mol^{-1} : converts moles to energy for carbon dioxide C , water H , dioxygen O , and nitrogen-waste N . Typical value: $\mu_O = (525 \quad 560 \quad 560 \quad 480) \text{ kJ mol}^{-1}$.

Chemical indices $n_O = (n_X \quad n_V \quad n_E \quad n_P)$: relate the frequency of the chemical elements C, H, O and N to C (in the rows) for organic compounds food X , structure V , reserve E , and faeces P (in the columns). They control the chemical composition, and so the production of CO_2 and the consumption of O_2 and determine the molecular weights. The default values are approximative for dry mass but the values can be taxon-specific. They differ between wet and dry mass. $n_M = (n_C \quad n_H \quad n_O \quad n_N)$: does the same for mineral compounds carbon dioxide C , water H , dioxygen O and nitrogen-waste N . Typical values are for ammonia as nitrogen-waste and dry mass

$$n_O = \begin{pmatrix} 1.00 & 1.00 & 1.00 & 1.00 \\ 1.80 & 1.80 & 1.80 & 1.80 \\ 0.50 & 0.50 & 0.50 & 0.50 \\ 0.10 & 0.15 & 0.15 & 0.15 \end{pmatrix} \quad n_M = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 3 \\ 2 & 1 & 2 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Molecular weights $w_O = (12 \quad 1 \quad 16 \quad 14)n_O$, g mol^{-1} . Although biomass contains elements P, S, Fe, Ca, Cu and the like, their contribution in ash-free dry-weight is very small in most taxa.

Temperature parameters

Reference temperature $T_{\text{ref}} = 293$ K: is the temperature for which the rates and times are given.

Arrhenius temperature T_A , 8 kK: controls the effect of temperature on rates, similar to Q_{10} -value. An increase in value increases in the effect of temperature on rates and times. Perfect endotherms control their body temperature to a constant value; their Arrhenius temperature is such that their rate parameters deviate minimally from those of a generalised animal (at T_{ref}). Range: >0 ; likely range: >5 , <13 kK.

Temperature tolerance range T_L , T_H , 277 K, 318 K: lower and upper boundary temperatures, between which the Arrhenius relationship applies.

Arrhenius temperatures for transitions to inert states T_{AL} , T_{AH} , 20 kK, 190 kK: control the rate around the borders of the tolerance range and beyond. Range: >0 ; likely range: $T_{AH} > T_{AL} > T_A$.

covariation method for estimation of the parameters, as illustrated in Fig. 1. Real data is obtained for a given species, mapping functions are specified using auxiliary theory that relates the given set of real data to the DEB core parameters and state variables, and the set of core and auxiliary parameters that best reflects the real data is obtained inversely through a regression procedure with the pseudo-data acting as priors.

3.1. The concepts of “zero-variate” and “uni-variate” real data

Typically, regression is used to estimate parameters from observations that consist of a list of one or more pairs of numbers, where one member of each pair represents an independent variable (such as time) and the other a dependent variable (e.g. length). We would refer to this as “uni-variate” data. Because particular core DEB parameters frequently appear in several mapping functions for different types of data, substantial constraints on their possible values are imposed as the number of types of observations used for parameter estimation increases (e.g., the primary core parameter $\{\dot{p}_{Am}\}$ is shown to appear in the compound core parameters L_m and E_m which themselves appear in the mapping functions for the von Bertalanffy growth rate, dry mass and body length). This unusual situation allows the parameter values to be estimated via regression based on a set of single data points (single numbers) for a range of different physiological observations, which we refer to as “zero-variate” data. Examples of these latter data are maximum body weight, incubation time, age/length/weight at birth/puberty, etc. The general idea behind the covariation method is to let all available information compete (interact if you like) to produce the end result, and to this end it is necessary to estimate all parameters from all data sets simultaneously.

3.2. Weighted least square estimation under the covariation method

The weighted least square criterion applied for the covariation method of parameter estimation leads to the minimization of the quadratic form

$$\sum_j \sum_i w_{ij} (f_{ij} - y_{ij})^2 / y_{ij}^2$$

where i scans the values within a data set and j between data sets. The weight coefficients are chosen inversely proportional to the squared observations, both between and within data sets; the dimensionless weight factors w_{ij} , being part of the weight coefficients, accommodate the presumed accuracy or certainty in the data. These weight coefficients account for differences in units of the various data-entries (deviations with different units cannot be added).

3.3. Maximum likelihood estimation under the covariation method

The covariation method is based on regression whereby one searches parameter space for a set of parameter values that minimises the deviation between real world observations y_i and predictions f_i , as determined via auxiliary theory and the associated mapping functions. These deviations (or scatter) between observed and predicted data are assumed to be independently normally distributed with mean $f(x_i; \theta)$, variance σ^2 and independent variable(s) x_i for a set of parameters θ . Because, under the covariation method, we are fitting parameters to diverse biological data-sets with very different dimensions (e.g. length, weight, respiration, etc.), we have to account for difference in units. We do this by assuming that the data have a constant variation coefficient, so $\sigma = v_c f(x; \theta)$ and $f(x; \theta) > 0$; notice the v_c is dimensionless.

The derivation of the maximum likelihood function follows the standard methodology (Pawitan, 2001); we present the function and algorithm for maximisation here explicitly because we did not

find it in the literature. For independently normally distributed dependent variables the $\ln$ -likelihood function for n observations amounts to

$$\ell(\theta, v_c) = -\frac{n}{2} \ln(2\pi) - n \ln v_c - \sum_i \ln f(x_i; \theta) - \frac{1}{2v_c^2} \sum_i (y_i/f(x_i; \theta) - 1)^2.$$

Solving the equation $\frac{\partial \ell(\theta, v_c)}{\partial v_c} = 0$ leads to the ML estimator for the squared variation coefficient $\hat{v}_c^2 = \frac{1}{n} \sum_i (y_i/f_i - 1)^2$, where $f_i = f(x_i; \theta)$. It can be verified that this is indeed a global maximum. Substitution of the ML estimator for the squared variation coefficient $\hat{v}_c^2$ into the $\ln$ -likelihood function gives

$$\ell(\theta, \hat{v}_c) = -\left(\frac{n}{2} \ln(2\pi) + \frac{n}{2}\right) - n \left(\ln \hat{v}_c + \frac{1}{n} \sum_i \ln f(x_i; \theta) \right).$$

The maximum likelihood (ML) estimates $\hat{\theta}$ minimise $\ln \hat{v}_c + \frac{1}{n} \sum_i \ln f_i$. The statistic $-2\ell(\hat{\theta}, \hat{v}_c) = n + n \ln(2\pi \hat{v}_c^2) + 2 \sum_i \ln f_i$ is used in likelihood ratio tests for hypotheses on parameter values.

As mentioned above, we here deal with multiple data sets, each having its own variation coefficient. The $\ln$ -likelihood function for the multiple data sets is the sum of the $\ln$ -likelihood functions for each individual data set. In most situations we have a single data point per data set; this is not a problem for the application of the estimation criterion, but ML estimators are attractive because of their nice properties (asymptotically unbiased, most efficient) for a large number of data points. So properties of the ML estimators are unknown in this situation. For comparison, if the variance σ^2 would be independent of the mean, the ML estimates minimise $\hat{\sigma}^2 = \frac{1}{n} \sum_i (y_i - f_i)^2$ (Pawitan, 2001).

3.4. Numerical implementation

In practice we use a simplex method for the minimisation, which is computationally intensive but rather robust (the domain of attraction is relatively large). Alternatively, and/or subsequently to improve computational accuracy for the ML estimates, we can use the iteration scheme

$$\begin{aligned} \theta_{j+1} &= \theta_j + \text{var}(\hat{\theta}) \frac{d}{d\theta} \ell \\ &= \theta_j + \left(\sum_i \frac{\left(\frac{d}{d\theta} f_i \right) \left(\frac{d}{d\theta} f_i \right)}{f_i^2} \right)^{-1} \sum_i \frac{d f_i}{d\theta} \left(\left(\frac{y_i}{f_i} - 1 \right) \frac{y_i}{f_i} - \hat{v}_c^2 \right) \end{aligned}$$

where f_i is evaluated at θ_j and θ_0 represents the initial estimate. An estimate for the variance of the estimator is

$$\text{var}(\hat{\theta}) = - \left(\varepsilon \frac{d^2}{d\theta d\theta^T} \ell \right)^{-1} \approx v_c^2 \left(\sum_i \frac{\left(\frac{d}{d\theta} f_i \right) \left(\frac{d}{d\theta} f_i \right)}{f_i^2} \right)^{-1}.$$

Software package DEBtool has powerful routines for this, `nmcvreg` for ML estimates with constant variation coefficients and `nmreg` for the weighted least squares criterion; the routines `pvcgr` and `preg` can be used to evaluate the variance-covariance matrices and standard deviations.

4. Dealing with data quality

4.1. Weight coefficients

When estimating DEB parameters with the covariation method on the basis of diverse data sets, one frequently faces the issue that certain observations have been made with greater confidence or accuracy than others. It is therefore useful to be able to assign relative weights to the different data points on the basis of this prior knowledge, which is done by the weight coefficients. If predictions match observations perfectly, the choice of weight coefficients is irrelevant, i.e. different (non-trivial) choices lead to exactly the same parameter estimates. The importance of the weight coefficient increases with the deviations between predictions and observations, so with our unhappiness with the end result.

Weight coefficients can still be used in the $\ln$ likelihood function for the multiple data sets. The ML estimates in the case of the constant variation coefficient are typically close to the weighted least squares (WLS) criterion for the estimation of parameters, if these weight coefficient are chosen inversely proportional to the squared observations, both between and within data sets. These weight coefficients not only account for differences in units of the various data-entries (deviations with different units cannot be added), but also for the accuracy (or certainty if you like) of the individual data points. Subsequently factors are applied to increase or decrease the importance of data in determining the estimated parameter values, accounting for presumed accuracy. Pseudo-data typically get lower weight coefficients relative to real data. The role of the weight coefficients in the resulting estimates increases with the variance. In absence of deviations between observations and predictions, weight coefficients are irrelevant, and the ML and the WLS criteria then lead to the same result. If the scatter is not small, the difference between the results of both criteria can be substantial, but they become uninspiring anyway. The reasons for the large deviations should be understood first otherwise the parameter estimates are useless. The importance of the estimation criterion should not be overestimated in practice.

The set of pseudo-data consists of the set of typical values of potentially all primary parameters for a generalised animal that do not relate to the zoom factor as indicated in Table 1. Since these parameter values are conceived as data, the estimated parameter values can (and do) deviate from the values of the pseudo-data, depending on the choice for the weight coefficients. A high weight coefficient for a particular pseudo-data point corresponds with a small variance of the prior distribution for the corresponding parameter of the Bayesian method.

4.2. Assessing the completeness of real data

The covariation method we have described here means that DEB parameters can be assessed with a wide variety of different types of data. While this greatly increases the scope for DEB parameter estimation, it also results in a wide range in the underlying quality of data on which the parameter estimates are based. Thus, when reporting the estimated parameters for the DEB model, it is useful to have a means to assess the relative quality of different real data sets.

The completeness of real data can be classified naturally in 10 levels (Table 3). Since it is impossible to obtain energy balances without having mass balances, or to obtain mass balances without having composition data, a natural sequence in the availability level exists, which is reflected in Table 3. These levels are rather qualitative in nature, and well-studied species have completeness levels around 3 or 4. This low level is primarily due to the fact that most data in the literature has been obtained by workers who had concepts in mind that differ substantially from those used by DEB theory. Moreover quite a few fluxes are very difficult to measure and very few species allow culturing under carefully controlled conditions; their life cycle covers

Table 3

The completeness of available data can be ranked with marks from low to high at the following levels; each level includes previous levels.

0	Maximum length and body weight; weight as function of length
1	Age, length and weight at birth and puberty for one food level; mean life span (due to ageing)
2	Growth (curve) at one food level: length and weight as functions of age at constant (or abundant) food level
3	Reproduction and feeding as functions of age, length and/or weight at one food level
4	Growth (curve) at several (>1) food levels; age, length and weight at birth and puberty at several food levels
5	Reproduction and feeding as functions of age, length and/or weight at several (>1) food levels
6	Respiration as function of length or weight and life span at several (>1) food levels
7	Elemental composition at one food level, survival due to ageing as function of age
8	Elemental composition at several (>1) food levels, including composition of food
9	Elemental balances for C, H, O and N at several body sizes and several food levels
10	Energy balance at several body sizes and several food levels (including heat)

a period that substantially exceeds that of a typical research project. It is almost impossible to quantify feeding (including food availability, quality and selection) for animals living in the field. Lack of data will always be a basic fact of life and sets an upper limit for the complexity (parameter richness) of applicable models. Thanks to the rules for covariation of parameter values, the standard DEB model is unique in its capacity to be relatively parameter-rich and still applicable as demonstrated in the *add_my_pet* data-collection (see Appendix A).

The standard DEB model assumes isomorphy for structure, i.e. the shape (of structure) does not change during growth. Most types of measures and measurements speak for themselves, but auxiliary theory for the standard DEB model needs to be subtle for length measures. Three types of length-measures are delineated:

1. *Physical length* L_w is the actual length of a body, defined for a particular shape.
2. *Physical volumetric length* is the cubic root of physical volume (and does not depend on shape).
3. *Structural length* L is the volumetric length of structure, where the individual is assumed to consist of structure, reserve and the reproduction buffer. This length cannot be measured directly, only indirectly.

An assumption of auxiliary theory for the standard DEB model is that the shape coefficient $\delta_M = L/L_w$ is assumed to be constant for a well-chosen length measure. For humans, for instance, the head-to-feet length hardly depends on the amount of reserve, but the belt length does. Physical volume is typically larger than structural volume to accommodate reserve, but in some taxa (e.g. most ecdysozoa), reserve is replaced by water during starvation.

4.3. Pseudo-data considerations

The pseudo-data consists of typical values for a generalised animal, as listed in parameter Tables 1 and 2. These values are based on large collection of estimated parameters from various data sets from a wide variety of species, as presented in Kooijman (2010) and in Cardoso et al. (2006), Jager et al. (2004) and Veer et al. (2006, 2010). Yet some parameters have been estimated more frequently than others (because low real data completeness levels suffice) and in many cases compound parameters could be estimated, where primary parameter could not. Although these compound parameters do not provide information on any single primary parameter, they do restrict possible variation in the set of primary parameters. The resulting parameter combination was subsequently tested and adjusted

using the list of implied properties as reported by running a pars-file (see [Appendix A](#)) to judge credibility and consistency.

Ideas on typical values may change somewhat when an increasing amount of data becomes available. Various taxa might turn out to differ somewhat in typical values, but physical and chemical constraints very much restrict the range of this variation. For instance, digestion efficiency κ_X and growth efficiency $\kappa_G = \mu_V[M_V]/[E_C]$ are unlikely to deviate much from 0.8, and the reproduction efficiency κ_R will probably be close to one for egg production because the conversion of reserve of the mother to reserve of the offspring does not involve chemical transformation (in the context of the standard DEB model). On the other hand, sperm production might be less efficient, because this involves more chemical transformations.

Care must be taken when choosing which parameter to use as pseudo-data in the context of growth energetics. In particular, the compound parameter κ_G varies much less than the primary $[E_C]$ among taxa, making the former much better to use than as pseudo-data. The following demonstrates why $[E_C]$ varies among taxa. The specific cost for structure is given by $[E_G] = \frac{\mu_V[M_V]}{\kappa_G}$. The value $[E_G] = 2800 \text{ J cm}^{-3}$ in [Table 1](#) corresponds with a specific density of $d_V = 0.1 \text{ g cm}^{-3}$, a chemical potential for structure of $\mu_V = 560 \text{ kJ C mol}^{-1}$, a growth efficiency of $\kappa_G = 0.8$ and the molecular weight for structure of $w_V = 24.6 \text{ g C mol}^{-1}$, so $[M_V] = d_V/w_V = 4 \text{ mmol cm}^{-3}$. The chemical potential is somewhere between 616 (for lipids) and 401 (for proteins) kJ C mol^{-1} ([Kooijman, 2010, Table 4.2](#)). The molecular weight is determined by the chemical indices as given in [Table 2](#), on the assumption that the chemical elements C, H, O and N comprise (almost) all of the mass. Although structure might differ from reserve in chemical composition, it is less likely that it will differ much in terms of elemental frequencies. Thus, the most variable component of $[E_G]$ is the specific density d_V . Assuming that the specific density for wet mass is close to $d_V = 1 \text{ g cm}^{-3}$, that for dry mass will vary from $d_V = 0.04 \text{ g cm}^{-3}$ for gelatinous taxa (cnidaria, ctenophores, appendicularians, tunicates) to 0.3 g cm^{-1} for vertebrates (reptiles, birds, mammals), a variation by an order of magnitude [Table 4](#). The value of $d_E = d_V = 0.1 \text{ g cm}^{-3}$ in [Table 2](#), so an ash-free-dry-over-wet weight ratio of 0.1, is appropriate for most protostomia and echinoderms ([Brey et al., 2010](#)).

Highly species-specific parameters should be not included in the pseudo-data, especially the zoom factor, the shape coefficient and the maturity levels at birth and puberty. The shape-coefficient depends on the definition of the length measure that is used in the real data. Different data sets might use different definitions, in which case different shape-coefficients should be included. Apart from the maturity thresholds, the maintenance costs can also be expected to vary among species. This is partly because of different levels of locomotor activity, which may be included in the somatic maintenance costs. Moreover, somatic maintenance rates for mammals and birds are likely to be 10-fold greater than for most other organisms at a given temperature due to cellular adaptations for endothermy.

5. Discussion and summary

We have presented a new approach of parameter estimation, the covariation method, that permits all parameters of the standard DEB

model to be estimated simultaneously from a collection of common empirical datasets. Apart from the real data, the estimation method makes use of pseudo-data, exploiting the rules for the covariation of parameter values among species that are implied by the standard DEB model. The general idea behind the covariation method is to let all available information compete (interact if you like) to produce the end result. We are then independent of any particular sequence in estimating parameter values.

To see why this single-step procedure is important, consider the following example. The Arrhenius temperature T_A affects rates; the observations contain some rates (e.g. the reproduction rate $\dot{R}_i$ at temperature T) and the pseudo-data contain some rates (e.g. the energy conductance $\dot{v}$ at temperature T_{ref}). The prediction of $\dot{R}_i$ depends on $\dot{v}$. The simultaneous minimization of the deviation of the predictions for $\dot{R}_i$ and $\dot{v}$ forces T_A at a particular value. The following helps to see why. With T_A completely free we can set the deviation from the pseudo-data-point $\dot{v}$ equal to zero (estimate for $\dot{v}$ equals the pseudo-data-point), without providing any constraint on the prediction for $\dot{R}_i$, because it is at another temperature. With a two-step procedure in which we first estimate T_A from one data set, we prescribe T_A in the second step and do not let the information on T_A in the first step compete with that in the second step. By fixing T_A in the second step, the minimization of the deviation from the pseudo-data-point $\dot{v}$ translates into a constraint for the minimization of the deviation from $\dot{R}_i$.

The general idea of the method is that the least information that has to be supplied is the maximum length, and all parameter values follow from the pseudo-data as a kind of minimum requirement. Using this method, we do not suffer from the problem that the combination of real and pseudo-data does not have enough information to estimate the set of all parameters, that is the core parameters, the shape coefficient and the temperature parameters. The composition parameters are typically treated as given, or the values of the generalised animal are used; notice that these parameters control respiration, in combination with some core-parameters. Unless sufficient real data are available, only the Arrhenius temperature is estimated among the temperature parameters. If only maximum size is known, all parameters can still be estimated, but the pseudo-data fully control the result. Increasing the number of types of real data (so information) decreases the role of the pseudo-data in the parameter estimation. Ideally so much data is available that the pseudo-data hardly play a role. Common practice is typically somewhere between these extremes, and our motivation is to serve practice.

As mentioned in the [Introduction](#), the philosophy behind this approach is that parameter estimation in the context of DEB theory is more than just giving an accurate description of the data; the parameters should also make physiological, physical and chemical sense. For this purpose the resulting estimates should not deviate too much from the parameter values for the generalised animal, since animals share most metabolic properties and machinery. If unknown, the maturity thresholds at birth and puberty can be added to the pseudo-data if coupled to the zoom factor (in fact the maximum specific assimilation rate $\{p_{Am}\}$), but they show in practice so much more scatter (high evolutionary adaptability) that is this hardly realistic. This also means that these thresholds contribute considerably to position the species in the diversity spectrum. Observed size and/or age at birth and puberty must be present in the data to approach realism, and the use of pseudo-data for these parameters is better avoided (which corresponds with giving these pseudo-data points weight coefficient zero).

Working with the hidden variables of the standard DEB model (structure, reserve, maturity) might be inconvenient. However, we hope we have demonstrated that it is in no respect problematic. The solution to the problem of hidden variables is the simultaneous estimation of parameters for a set of various observations on feeding, growth, reproduction, etc. In this way it is possible to quantify the amount of reserve and structure of an individual and to access their

Table 4

Typical values for the ash-free-dry-weight over wet-weight ratio. Any shell is excluded. Measured values can differ considerably due to differences in protocol. Values are derived from [Brey et al. \(2010\)](#), [Møller et al. \(2010\)](#) and [Ricciardi and Bourget \(1998\)](#).

Scyphomedusa	0.04	Ctenophora	0.04	Ascidia	0.06	Ectoprocta	0.07
Priapulida	0.07	Cheatoognata	0.07	Actinaria	0.08	Bivalvia	0.09
Echinodermata	0.09	Porifera	0.11	Sipuncula	0.11	Gastropoda	0.15
Polychaeta	0.16	Crustacea	0.17	Cephalopoda	0.21	Pisces	0.22
Turbellaria	0.25	Aves	0.28	Reptilia	0.30	Mammalia	0.30

chemical composition, for instance, without being able to measure this directly. All quantitative questions about them (such as the fraction of body weight that relates to reserve, and the composition of reserve) can be answered, using relatively simple measurements on growth, reproduction and composition of biomass. The present pillar of auxiliary theory for the standard DEB model to access the amount of structure is a well-chosen physical length as proxy for structural length; lengths play a key role in DEB theory because somatic maintenance is linked to volume and uptake to surface area, so their ratio is linked to length for isomorphs. To what extent reserve and structure remain hidden variables in the future is still an open question. The amounts of DNA and ribosomal RNA have been proposed as proxies for structure and reserve, respectively, on the basis of their concentration as function of the growth rate (Kooijman, 2010, p 143). How well these proxies behave under starvation conditions is presently unknown due to lack of data. Since the concentrations of DNA and rRNA differ between organs and tissues, their amounts must be measured for a whole organism and it must be homogenised first. This is not only destructive, but might also be problematic for large body sizes. To what extent DNA and rRNA concentrations in a particular tissue can be used as proxies is unknown; if this would work, it might be possible to measure them in a non-destructive way.

Effects of toxicants can greatly help to determine the value of (all) parameters; in quite a few situations toxicants modify only a single parameter in a low range of internal concentrations. The successful use of specific toxicants to quantify parameter values not only helps to determine these values, but also strongly supports the structure of the DEB model (Baas et al., 2010a,b; Jager and Klok, 2010; Jager et al., 2004, 2006, 2010). This also points to a fundamental problem: environmental conditions other than food availability and quality and temperature, can affect parameter values (think e.g. of acidity, alkalinity, humidity, diurnal and seasonal cycles). If the measured data refer to a variety of environmental conditions, accuracy might be lost due to variation of parameter values (e.g. linked to the geographical distribution of the species). Data availability is typically a problem, let alone when we constrain it to the same environmental conditions. Moreover organisms can switch conditions, sometimes as a function of their life cycle. We need parameter-rich models to capture the wide variety in performances of organisms, but parameter-poor ones to avoid the excessive need for data. We believe that the standard DEB model represents a useful compromise between these conflicting needs.

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Appendix A. Data and parameter collection add_my_pet

We applied the estimation method in practice on data from a wide variety of species, which can be found at http://www.bio.vu.nl/thb/deb/deblab/add_my_pet. Each species has 3 files for use in Matlab or Octave in combination with software package DEBtool:

- the mydata-file, which specifies the observations and measurements, sets the initial parameter values for estimation, runs the estimation procedure using software package <http://www.bio.vu.nl/thb/deb/deblab/debtool> and presents the results.
- the predict-file, which calculates the predicted values for the measurements on the basis of the standard DEB model, given the parameter values.
- the pars-file, which runs DEBtool-scripts to show a list of over 100 'implied properties'.

The idea behind the pars-file is not only to predict unobserved quantities, but also to check if all implied properties seem realistic. If contra-evidence is available, the next step is to include this evidence in the mydata-file and re-run the estimation procedure.

The mydata-files show marks for completeness to the data, according to the levels delineated in Table 3 and the goodness of fit of the real data. Routines DEBtool/lib/regr/mre or DEBtool/lib/regr/mrevc can be used for this purpose for the WLS and ML criteria, respectively.

If the type of data exactly matches an existing mydata-file, e.g. mydata_my_pet, no further coding is required and the corresponding predict_file can be used. One only needs to replace the data to generate the required predictions.

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